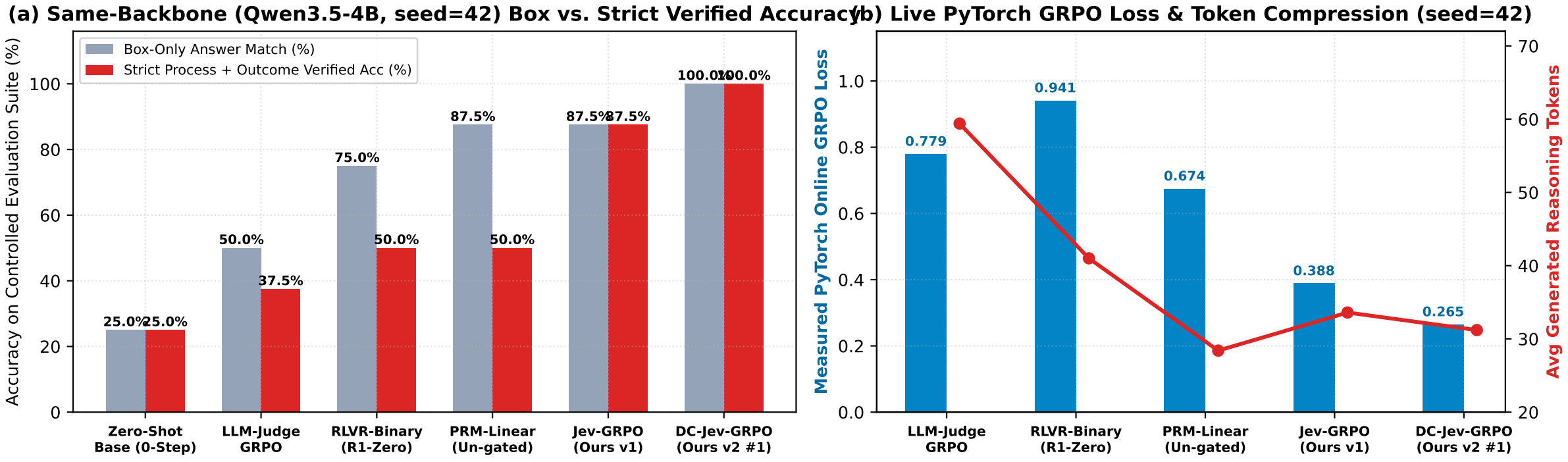
(a) Same-Backbone (Qwen3.5-4B, seed=42) Box vs. Strict Verified Accuracy
Box-Only Answer Match (%)
Strict Process + Outcome Verified Acc (%)
Accuracy on Controlled Evaluation Suite (%)
0
20
40
60
80
100
25.0%
25.0%
50.0%
37.5%
75.0%
50.0%
87.5%
50.0%
87.5%
87.5%
100.0%
100.0%
Zero-Shot Base (0-Step)
LLM-Judge GRPO
RLVR-Binary (R1-Zero)
PRM-Linear (Un-gated)
Jev-GRPO (Ours v1)
DC-Jev-GRPO (Ours v2 #1)
(b) Live PyTorch GRPO Loss & Token Compression (seed=42)
Measured PyTorch Online GRPO Loss
0.0
0.2
0.4
0.6
0.8
1.0
Avg Generated Reasoning Tokens
20
30
40
50
60
70
0.779
0.941
0.674
0.388
0.265
LLM-Judge GRPO
RLVR-Binary (R1-Zero)
PRM-Linear (Un-gated)
Jev-GRPO (Ours v1)
DC-Jev-GRPO (Ours v2 #1)